

7. Two samples of modelling clay, X and Y, have the same shape and mass.

Sample X is made of a softer clay than sample Y.

The samples are both released from the same height and fall towards the ground.

Both samples come to rest when they collide with the ground.

A student makes the following statements about the samples colliding with the ground.

- I Sample X comes to rest in less time than sample Y.
- II The mean force exerted by the ground on sample X is less than the mean force exerted by the ground on sample Y.
- III The change in momentum of sample X is less than the change in momentum of sample Y.

Which of these statements is/are correct?

- A I only
- B II only
- C I and II only
- D I and III only
- E II and III only